

4	
(a)(i)	A <i>polarised wave</i> is a wave whose vibrations are restricted to one plane.
(a)(ii)	Apply Malus' Law, $I_1 = I_0(\cos 40^\circ)^2$
	$I_2 = I_0(\cos 40^\circ)^2(\cos 80^\circ)^2$ $= 0.018I \text{ (2 s.f.)}$
(b)(i)1.	Displacement of wave laterally (left and right) from the central axis of the tube represents vibrations of air molecules vertically (up and down) along the axis of the tube.
	The vibrations are parallel to the direction of propagation of the incident wave down the tube.

(b)(i)2.	Correct stationary wave pattern.
(b)(i)3.	Correct number of nodes and location of nodes.
(b)(ii)	$\frac{1}{2} \lambda = 35.0 \text{ cm}$ $\lambda = 70.0 \text{ cm}$
	$f = \frac{340}{0.700}$ $= 486 \text{ Hz (3 s.f.)}$
(c)(i)	$d \sin \theta = n \lambda$ $\frac{1}{5.00 \times 10^5} \sin 90^\circ = n(540 \times 10^{-9})$
	$n = 3.70$ $= 3 \text{ (round down to nearest whole number)}$
	$\text{Total number of maxima} = (3 \times 2) + 1$ $= 7$
(c)(ii)	$d \sin \theta = n \lambda$ $\frac{1}{5.00 \times 10^5} \sin \theta = 2(542 \times 10^{-9})$ $\theta = 32.820^\circ$ $d \sin \theta = n \lambda$ $\frac{1}{5.00 \times 10^5} \sin \theta = 2(539 \times 10^{-9})$ $\theta = 32.616^\circ$
	Angular separation is 0.204° Since angular separation is less than 0.30°, the two wavelengths cannot be distinguished at the second order.
E	